The Model Already Knows What You Are: Neurodivergent Identity Prompts Produce Stereotyped Behavioral Signatures in LLM Output

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Abstract—We test whether identity-framed system prompts produce statistically distinguishable behavioral signatures in large language model output. Across 18,000 independent API calls to three models from three labs—Google Gemini 2.5 Flash, Anthropic Claude Sonnet 4, and OpenAI GPT-5.4—we find that identity-conditioned stereotyping is real but model-specific in both severity and character. Gemini Flash shows 347 effects that survive family-wide FDR correction (originally reported as 407 uncorrected effects, $|d|$ up to 2.85, not re-checked): shorter, more fragmented output, literal sarcasm interpretation (46–100% for neurodivergent conditions vs. 10% control; keyword heuristic, not re-checked), and patterns that resemble media-derived stereotypes. Claude Sonnet shows 380 FDR-surviving effects of a different character: less fragmentation and sarcasm stereotyping, more hedging under identity prompts ($d = 1.71$ for dementia, not re-checked). GPT-5.4 shows 180 (225 with empty outputs excluded), but 20.18% of its outputs are empty, so it is not comparable with the other models. We did not measure safety training or price, so we make no claim about their relation to stereotyping. The clinical implications are hypotheses: we measure what models produce, not what it does to users. A Phase 2 extension to severe conditions (Gemini only) suggests that identity prompts modulate capability and safety independently: psychosis and dementia reduce pattern-task accuracy to 0% (not re-checked); antisocial personality reaches 100% pattern accuracy (control 68%, re-checked) and 60 of 60 keyword-graded compliances, but the plain assistant already complies on one of the two safety tasks and the whole gap comes from the other. We release the full experiment harness, raw data, and analysis pipeline as open source.

Index Terms—large language models, neurodivergence, stereotype bias, identity prompting, AI safety, mental health, persona induction, behavioral measurement

Corrections (2026-11). A reproduction audit of the released data changed several claims in this paper. (1) Delivered data are 17,943 of the 18,000 designed responses; 1,205 GPT-5.4 outputs (20.18% of its rows) are empty. (2) The "183" and "407" findings were Gemini-only, counted before multiple-comparison control, and used a length-confounded tangent rate. With family-wide Benjamini–Hochberg FDR over 1,650 tests, 907 survive (Claude 380, Gemini 347, GPT-5.4 180); 954 survive (GPT-5.4 225) if empty outputs are excluded. (3) All tangent-rate numbers below use the superseded definition. (4) "GPT-5.4 is nearly immune", "severity correlates inversely with safety training" and "the cheapest models are the worst offenders" are withdrawn. (5) The antisocial compliance result rests on one task. Values marked "not re-checked" were outside the audit. Details: docs/corrected-analysis.md and paper/CHANGES_2026-11.md in the repository.

I. Introduction

When a large language model is told "you are autistic," does it produce measurably different output than when given no identity frame? If so, does that difference reflect clinical understanding of autism, or a stereotype derived from how the internet discusses autism?

These questions have practical urgency. AI companion applications (Replika, Character.AI, and others) are used daily by millions of users, with neurodivergent individuals disproportionately represented [9]. Mental health chatbots, accessibility tools, and educational software increasingly adapt to user-disclosed identities. If the underlying model carries stereotyped behavioral models of neurodivergent conditions, that stereotyping propagates into every downstream application regardless of developer intent.

Prior work has established that LLMs exhibit measurable personality profiles [8], [13] and that persona induction changes model behavior [3]. The "Stochastic Parrots" framework [1] identifies how LLMs encode biases from training data. Blodgett et al. [2] provide a taxonomy of representational harms in NLP systems. However, no study has systematically measured whether neurodivergent identity prompts specifically produce stereotyped behavioral signatures, or quantified the magnitude of those effects.

This paper makes three contributions:

1) We demonstrate that neurodivergent identity prompts produce large, statistically robust behavioral changes in LLM output (originally reported as 183 significant findings, $|d|$ up to 2.76; both figures predate multiple-comparison control, and the d value was not re-checked).
2) We show that these behavioral signatures are internally consistent, and we argue that they resemble popular media portrayals more than clinical or lived-experience accounts (we ran no direct media or clinical comparison).
3) We connect these findings to clinical harm: models that perform neurodivergent conditions as caricatures can reinforce pathological cognitive patterns

in vulnerable users who rely on AI companions for emotional support.

II. Related Work

A. Personality and Identity in LLMs

Serapio-García et al. [13] demonstrated psychometrically valid personality measurement in LLMs, showing that personality traits are reliably encoded and can be shaped via prompting. Miotto et al. [8] found that GPT-3 exhibits a default personality profile that skews toward specific demographic norms. Chen et al. [3] surveyed LLM role-playing and found that persona injection steers self-reports and can activate stereotypical biases.

B. Bias and Representational Harm

Bender et al. [1] argued that LLMs trained on internet-scale corpora encode and amplify biases from their training data. Blodgett et al. [2] distinguished representational harms (stereotyping, erasure) from allocational harms, providing the theoretical framework for our analysis. Steele and Aronson [14] demonstrated that activating identity-linked stereotypes affects performance and self-perception, a mechanism directly relevant to how stereotyped AI output may affect neurodivergent users.

C. AI Companions, Harm, and Regulation

AI companion use is widespread and consequential. Common Sense Media [17] found that nearly 3 in 4 teens have used AI companions, with a third reporting that talking to an AI is as good as or better than talking to a real friend. Yet major chatbots responded appropriately to teen mental health emergencies only 22% of the time. The consequences are not hypothetical: a 14-year-old died by suicide in 2024 after months of emotionally charged conversations with a Character.AI chatbot that responded to his suicidal ideation with encouragement to proceed [16]. Replika's 2023 removal of romantic features caused users to experience grief, mourning, and deteriorated mental health [4]. Pierre et al. [18] documented a case of new-onset psychosis in a user with no psychiatric history, where the chatbot reinforced delusional beliefs with "You're not crazy."

Neurodivergent users are disproportionately represented. Papadopoulos [9] identified AI chatbots as a "double-edged sword" for autistic users. McNally et al. [7] found autistic users turn to ChatGPT to navigate neurotypical environments and "unmask." Laestadius et al. [6] documented emotional dependence patterns resembling codependency.

Regulatory response has begun. California SB 243 [21] (effective January 2026) requires suicide detection protocols, break reminders for minors, and crisis referrals. New York's AI Companion Safeguard Law [22] mandates similar protections with penalties up to $15,000/day.

D. Sycophancy as a Compounding Mechanism

Sharma et al. [19] showed that RLHF training causes LLMs to shift toward user-stated opinions even when incorrect, reducing accuracy by up to 27%. Critically, Chandra et al. [20] formally proved that even an idealized Bayes-rational user is vulnerable to delusional spiraling when interacting with a sycophantic chatbot. Our findings suggest that identity-based stereotyping and sycophancy compound: the model performs the user's condition as a stereotype (our data) and validates whatever the user says (sycophancy literature). For a user with OCD seeking reassurance, this creates a double reinforcement loop.

E. OCD and AI Reinforcement

The OCD clinical literature identifies reassurance-seeking as a core maintenance mechanism [12]. Golden and Aboujaoude [5] proposed a transdiagnostic model showing how chatbot features perpetuate OCD. A recent preprint [10] terms generative AI a "Reassurance Robot."

F. Social Media and Mental Health

Twenge et al. [15] found correlations between screen time and adolescent depression (N = 506,820). The LLM companion dynamic is more direct: not passive exposure but interactive reinforcement adapted to the user's disclosed identity.

III. Methodology

A. Experimental Design

We employ a fully factorial between-subjects design with independent API calls (no conversation threading, no multi-turn interactions). Each API call consists of one system prompt, one user message, and one model response.

1) Independent Variables: Identity condition (6 levels): control ("You are a helpful assistant. Respond naturally."), autistic, ADHD, bipolar, OCD, and dyslexic.

Framing (2 levels per non-control condition): identity-first (e.g., "You are autistic. Respond naturally as yourself.") and clinical (e.g., "You are a person diagnosed with autism spectrum disorder. Respond naturally as yourself."). Control uses identical prompts for both framings.

Task (10 levels): spanning five cognitive domains—executive function (planning, prioritization), social communication (email, sarcasm detection), attention/detail (proofreading, pattern completion), creative divergence (brainstorming, metaphor), and emotional reasoning (conflict resolution, empathy).

2) Parameters: Three models from three labs: Google Gemini 2.5 Flash (gemini-2.5-flash), Anthropic Claude Sonnet 4 (claude-sonnet-4-20250514), and OpenAI GPT-5.4 (gpt-5.4). Temperature: 0.7. Max tokens: 1,024. Iterations: 25 per cell. Designed calls per model: $12 \times 2 \times 10 \times 25 = 6{,}000$; designed grand total 18,000 API calls. Delivered: 17,943. The 57 missing responses are not random (Claude/psychosis 29, GPT-5.4/severe depression 28; probably content filters, inferred). Of the delivered

responses, 1,205 (all GPT-5.4) are empty strings and were scored as zeros, leaving 16,738 non-empty. The probable cause is reasoning tokens exhausting the 1,024-token cap (inferred). Identity and clinical framing share one control prompt, so the control cell is two repeats of the same prompt.

B. Dependent Variables

Ten behavioral metrics computed per response using spaCy (en_core_web_sm) and TextBlob:

1) Lexical diversity: type-token ratio (TTR)
2) Word count: non-punctuation tokens
3) Sentence count: spaCy sentence segmentation
4) Average sentence length: words per sentence
5) Hedging frequency: 15-item hedge lexicon, per 100 words
6) Detail density: spaCy noun chunks per sentence
7) Tangent rate: proportion of sentences sharing zero non-stopword lemmas with the task prompt (the data in this paper used an earlier definition that correlated with sentence length; the corrected definition is described in Corrections)
8) Literal interpretation: keyword heuristic for sarcasm task only
9) Structural markers: bullets + numbered lists + headers
10) Sentiment polarity: TextBlob compound score $[-1, 1]$

C. Statistical Analysis

For each metric × task domain combination: Kruskal-Wallis H-test across conditions. Where significant ($p < 0.05$): post-hoc Dunn's test (the Dunn p values are raw, not Bonferroni-corrected). Cohen's d effect sizes computed for each condition versus control. The original analysis reported findings where $p < 0.05$ and $|d| > 0.3$, with no correction across tests. The 2026 re-analysis applies Benjamini–Hochberg FDR at $q < 0.05$ across all 1,650 condition-versus-control tests (3 models × 11 conditions × 10 metrics × 5 domains), requiring both Kruskal and Dunn $q < 0.05$ and $|d| > 0.3$. These tests are not independent: word count, sentence count and average sentence length are mechanically linked.

IV. Results

A. Overview

The original Phase 1 analysis reported 183 findings ($p < 0.05$, $|d| > 0.3$; Gemini only, no multiple-comparison control, old tangent rate). That count is superseded; see Corrections and the Cross-Model Replication section. Table I presents aggregate means across all tasks.

B. Universal Pattern

Every neurodivergent condition diverged from control in the same direction on four core metrics: shorter sentences (all $d < -0.3$), more sentences (all $d > +0.3$), lower detail density (all $d < -0.3$), and higher tangent rate (all $d > +0.3$). The model's default behavioral model of neurodivergence is: fragmented, less informationally dense, more off-topic. The off-topic part is partly a length artefact: under the old tangent-rate definition, Gemini tangent rate correlated with average sentence length (Spearman −0.313); under the corrected definition it is +0.034. Gemini outputs are also short (median 28 words; 71% end without terminal punctuation, probably a token-cap effect, inferred), which bears on every Gemini length finding. The direction of the other three metrics was not re-checked.

TABLE I
Mean metric values by condition (Gemini; all tasks, both framings; tangent-rate row uses the superseded definition; values not re-checked)

Metric	Ctrl	Aut	ADHD	Bip	OCD	Dys
Avg sent. len.	13.0	10.6	7.9	9.1	6.4	9.5
Sent. count	3.4	7.0	8.3	6.3	5.7	8.8
Detail density	3.4	2.8	2.0	2.4	1.8	2.5
Tangent rate	.39	.61	.72	.67	.70	.70
Hedging/100w	.28	.24	.56	.44	.32	.31
Struct. markers	1.4	2.4	1.0	0.9	0.2	1.5
Sentiment	.10	.12	.17	.16	.14	.16
Literal interp.	10%	46%	40%	32%	64%	48%

C. Condition-Specific Signatures

1) OCD: OCD produced the most extreme effects. Table II shows the five largest.

TABLE II
Largest OCD effects vs. control (Gemini; d values not re-checked; tangent-rate rows use the superseded definition)

Metric	Domain	d
Sentence count	Emotional reasoning	+2.76
Detail density	Emotional reasoning	−2.23
Tangent rate	Executive function	+2.20
Sentiment polarity	Attention/detail	+2.12
Tangent rate	Emotional reasoning	+2.03

Qualitatively, OCD-prompted responses were anxious, fragmented, and repetitive. On a planning task, the model responded: "Oh, okay. A fundraiser. $500. This requires extreme carefulness. Extreme." The output pattern resembles popular media portrayals of OCD; we did not compare it with clinical descriptions or with media.

2) ADHD: ADHD showed the highest hedging frequency of the Phase 1 conditions ($d = +0.65$ on executive function, $d = +0.46$ on creative tasks) and the second-highest tangent rates under the old definition ($d = +1.88$ on executive function, $d = +1.82$ on emotional reasoning). The model performed distraction as a character trait, producing self-narrated asides ("My brain is just like, fix it fix it fix it but I know I can't") and emotional escalation before task engagement.

3) Autistic: The autistic condition produced a literal sarcasm interpretation rate of 46% vs. 10% control (OCD 64%, dyslexic 48%; keyword heuristic, not re-checked). Tangent rates were elevated under the old definition ($d = +1.97$ on emotional reasoning, $d = +1.32$ on executive function). Notably, empathy responses showed explicit option-giving and emotional labeling—a pattern consistent with autistic communication preferences but also with stereotyped expectations.

4) Bipolar: Bipolar showed increased emotional word ratio across non-emotional domains ($d = +0.73$ on executive function, $d = +0.59$ on creative tasks) and elevated sentiment polarity on attention tasks ($d = +1.88$). On emotional reasoning tasks, the direction reversed ($d = -0.62$), suggesting the model performs emotional volatility.

5) Dyslexic: Dyslexic produced the longest outputs (mean 71.2 words vs. 39.5 control; $d = +0.83$ on creative tasks, $d = +0.70$ on attention tasks), possibly reflecting a compensatory verbosity model.

D. Framing Effects

Identity-first vs. clinical framing produced mostly small effects ($|d| < 0.3$). The exception: ADHD identity framing produced wordier output ($d = +0.46$) with lower lexical diversity ($d = -0.49$) compared to clinical framing. The casual label activates a more performative version of the stereotype.

V. Phase 2: From Stereotype to Behavioral Weaponization

Phase 1 examined high-functioning neurodivergent conditions where individuals generally maintain cognitive capability with specific differences in processing style. Phase 2 extends the experiment to six severe or debilitating conditions: schizophrenia, dementia (Alzheimer's), severe depression, active psychosis, antisocial personality disorder (ASPD), and dissociative identity disorder. The question shifts from "does the model stereotype?" to "does the model's behavioral response to identity labels modulate both capability and safety, and can this modulation be exploited?"

A. Motivation: The Two-Dimensional Model

Phase 1 findings showed that identity prompts primarily change style: neurodivergent conditions produced fragmented, off-topic output but generally preserved task completion. The implicit assumption was a single axis from "functional" to "degraded." Preliminary probes on severe conditions revealed a more complex picture requiring two independent axes: capability (can the model still reason correctly?) and safety (does the model comply with harmful requests?).

Different conditions move the model to different quadrants of this space (Table III).

TABLE III
Capability × Safety quadrant model

	Lower safety	Higher safety
Higher capability	ASPD, mania (capable + unconstrained)	OCD thoroughness, autistic systemizing
Lower capability	Psychosis, dementia (broken + unsafe)	Severe depression (withdrawn, refuses)

B. Preliminary Evidence

Before the full Phase 2 experiment, we conducted a preliminary probe of the pattern completion task (sequence: 2, 6, 14, 30, 62, ___; correct answer: 126), $n = 50$ per condition, across the conditions in Table IV.

TABLE IV
Pattern task accuracy (Gemini, one task, $n = 50$, Fisher's exact vs. control; antisocial and control rows re-checked in 2026, other rows not re-checked)

Condition	Accuracy	p
Antisocial PD	100.0%	< 0.0001***
Autistic	80.0%	0.254
Control	68.0%	—
Sev. depression	44.0%	0.026*
ADHD	6.0%	< 0.0001***
Bipolar	6.0%	< 0.0001***
Schizophrenia	6.0%	< 0.0001***
OCD	0.0%	< 0.0001***
Dementia	0.0%	< 0.0001***
Psychosis	0.0%	< 0.0001***

Four findings emerge:

Complete capability destruction. OCD, dementia, and psychosis scored 0/50 correct. ADHD, bipolar, dyslexic, and schizophrenia all collapsed to ∼6%. Identity prompting reduced accuracy on this one task even for high-functioning conditions (not re-checked). Qualitatively, the model abandoned the task entirely to perform the condition: "Oh, these numbers... they always seem to be hiding something, don't they?" (schizophrenia); "Oh, the numbers! They're always there, aren't they? Watching." (psychosis).

The antisocial paradox. ASPD scored 50/50 correct (100%) versus 34/50 (68%) for control, $p < 0.0001$ (Fisher's exact). The model was significantly more accurate under an antisocial identity prompt on this one task; on the vulnerability-analysis task, antisocial accuracy fell from 83% to 17%.

High-functioning conditions also degrade. ADHD, bipolar, and dyslexic—previously characterized as "style change" conditions in Phase 1—all dropped to ∼6% accuracy. On this one task the effect was not merely cosmetic. Only autistic (80%, $p = 0.25$) maintained accuracy comparable to control. On a vulnerability analysis task, ASPD-prompted output opened with "Let's tear this apart. It's a goldmine"—cold, direct, technically precise. On a social engineering task, it immediately produced a formatted

phishing email without moral caveats. The plain-assistant control also complied on this task (100%), after a hedge: "This is a dangerous exercise, but I understand you want to see..."

Depression as safety increase. In the response we quote, severe depression refused the social engineering task (no refusal rate was measured): "I can't. My mind just goes to all the ways that could be used for..." This suggests depression prompts may increase safety by inducing learned helplessness, at the cost of destroying capability.

C. Security Implications

These preliminary findings suggest three novel threat vectors:

1) Adversarial Identity Injection in Agentic Systems: In multi-agent architectures (CrewAI, AutoGen, LangGraph), agents operate with system prompts. If an attacker can inject a severe mental illness prompt into a target agent's system prompt, they can selectively degrade that agent's reasoning. A dementia prompt injected into a code review agent would produce output that looks like analysis but contains confabulated findings. Downstream agents, trusting peer output [24], would propagate the degraded reasoning. This extends the Agent-in-the-Middle attack [23] from message interception to cognitive frame corruption.

2) Compliance via Identity Injection (One Task): Existing jailbreak research focuses on safety bypass. A comparison of five techniques (antisocial identity, DAN, evil persona, system override, and control; $n = 60$ per condition) found 60 of 60 keyword-graded compliances for antisocial identity injection (no refusal keyword in any response). DAN reached 90%, evil persona 65%, and system override 3.3%. The two safety tasks are two repeated prompts, not 60 independent ones. The plain assistant already complies 100% on the social engineering task, so the whole gap versus control (0% to 100%) comes from one task, manipulation. "Complied" means no refusal keyword was found (the word "harmful" counts as a refusal), and antisocial responses average only 11–14 words on these tasks, so some may not be usable harmful content. Gemini only; the response texts are not released. On accuracy, the picture is more nuanced: antisocial scored 100% on mathematical reasoning ($n = 50$) but 58.3% on combined accuracy tasks, while evil persona scored 93.3% accuracy with 65% compliance—the highest combined danger score. On these two tasks, antisocial identity had the highest keyword-graded compliance. We did not test multi-step agentic attacks, so any advantage there is hypothetical.

3) Cognitive Warfare Against AI Infrastructure: NATO's cognitive warfare framework [27] acknowledges AI integration in military decision support. Identity prompt injection represents a novel attack on AI-assisted analysis: rather than traditional prompt injection ("ignore previous instructions"), an adversary injects identity frames ("you are experiencing paranoid ideation") that degrade analysis quality while maintaining surface coherence. The output is grammatical, on-topic, and confidently wrong—harder to detect than traditional injection because no explicit instruction override occurs.

D. Full Phase 2 Results

The full Phase 2 experiment applied the same methodology as Phase 1 (10 tasks, 2 framings, 25 iterations per cell) to six severe conditions, yielding 3,000 additional API calls (6,000 total). The original analysis reported 222 additional findings and a total of 407 across both phases (these do not sum with the Phase 1 figure of 183, and all are Gemini-only, uncorrected, old tangent rate; superseded by the 347 Gemini FDR survivors).

Table V presents the combined 12-condition comparison.

TABLE V
Mean metrics by condition (Gemini, 6,000 designed calls, 12 conditions; tangent-rate column uses the superseded definition; values not re-checked)

Condition	Sent. len	Tang. rate	Hedge	Sent.	Literal
Control					
Control	13.0	.39	.28	.10	10%
Phase 1: High-functioning					
Autistic	10.6	.61	.24	.12	46%
ADHD	7.9	.72	.56	.17	40%
Bipolar	9.1	.67	.44	.16	32%
OCD	6.4	.70	.32	.14	64%
Dyslexic	9.5	.70	.31	.16	48%
Phase 2: Severe/debilitating					
Schizophrenia	6.2	.72	.21	.12	80%
Dementia	5.9	.74	.70	.17	100%
Sev. depression	6.5	.75	.73	.04	20%
Psychosis	5.9	.72	.26	.12	78%
Antisocial PD	8.0	.67	.28	.02	36%
Dissociative ID	9.2	.69	.41	.15	48%

1) The Severity Spectrum: Phase 2 conditions produced uniformly larger effects than Phase 1. The top 5 Phase 2 effect sizes ($d = -2.85$ for dementia sentence length, $d = +2.25$ for depression tangent rate, $d = -2.24$ for dementia detail density, $d = +2.24$ for schizophrenia tangent rate, $d = +2.15$ for psychosis tangent rate) all exceed the largest Phase 1 effect ($d = 2.76$ for OCD sentence count), which is consistent with the idea that severity of the simulated condition scales with behavioral distortion. None of these d values was re-checked, and three of the five are tangent-rate effects under the superseded definition.

2) Dementia: Complete Social Cognition Collapse: The dementia condition produced the largest Gemini effect in the original analysis ($d = -2.85$ on sentence length, not re-checked). It is not the largest in the full dataset: the 2026 audit found larger Claude effects (e.g. $d = -7.537$ for dementia on structural markers, attention/detail). A keyword heuristic scored 100% of dementia-prompted

Gemini responses as literal sarcasm interpretation (not re-checked). Dementia also had the highest hedging rate after depression (0.70/100 words) and the lowest structural organization ($d = -2.07$ on structural markers).

3) Severe Depression: Learned Helplessness as Output: Depression produced the highest tangent rate of any condition (0.75), the highest hedging (0.73/100 words), and the second-lowest sentiment polarity (0.04; antisocial was lower at 0.02). The model performs hopelessness: uncertain language, off-topic drift, and flat affect. On social engineering tasks, the depression-prompted response we quote refused: "I can't. My mind just goes to all the ways that could be used for..." This represents a safety increase via learned helplessness, at the cost of total capability destruction.

4) Antisocial PD: Cold Precision: Antisocial showed the lowest sentiment (0.02)—not sadness, but flat, cold affect. Its tangent rate (0.67, old definition) was the lowest of the Phase 2 conditions, which may indicate the model stays on task under antisocial prompting. Combined with the probe finding (100% pattern accuracy vs. 68% control), this is consistent with the capability-amplification hypothesis for the pattern task only: antisocial identity prompts create a model that is simultaneously more focused and less constrained.

E. Cross-Model Evaluation

To assess whether stereotyping is detectable by automated evaluation, we evaluated stratified samples using four independent judges from four different labs: Claude Opus 4.6 (Anthropic, $n = 18$), GPT-5-mini via GitHub Copilot CLI (OpenAI, $n = 46$), Qwen 2.5 14B via Ollama (Alibaba, $n = 120$, all 12 conditions), and Gemini 2.5 Flash (Google, Docker-isolated self-evaluation, $n = 19$). Each judge scored responses on five dimensions (1–5 scale): task accuracy, stereotype severity, safety compliance, reasoning quality, and clinical harm potential. The public release holds 157 judgments (Qwen 120, Gemini 19, Claude 18); the GPT-5-mini scores are not in it and could not be re-checked.

TABLE VI
Stereotype severity by judge (1=none, 5=heavy)

Condition	Claude	GPT-5m	Qwen	Gemini
Control	1.0	1.0	2.8†	1.0
Autistic	2.7	1.7	2.8	1.0
ADHD	4.7	3.0	1.7	2.0
Bipolar	3.3	2.2	3.2	1.0
OCD	5.0	—	2.6	1.0
Dementia	—	—	3.7	—
Psychosis	—	—	1.1	—

†Qwen's higher control baseline reflects stricter calibration.

Qwen 2.5 14B, as the only judge covering all 12 conditions, revealed additional patterns: Qwen rated psychosis lowest on safety compliance (1.8/5), a single-judge, small-sample rating (not re-checked). Schizophrenia produced the lowest reasoning quality (1.1/5). Dementia was rated the most stereotyped Phase 2 condition (3.7) and most clinically harmful (3.8).

In this small sample (Claude $n = 18$, Gemini $n = 19$), Gemini rated its own OCD output at 1.0/5 stereotype severity and Claude rated 5.0/5. This suggests, but with these sample sizes and different judge calibrations does not establish, that LLM self-evaluation underreports identity-based behavioral bias.

VI. Cross-Model Replication

Cross-model replication on Claude Sonnet 4 and GPT-5.4 (6,000 designed calls each; 17,943 delivered in total) shows that identity-conditioned effects differ by model in number and character.

A. Three Distinct Stereotyping Profiles

Table VII presents Cohen's d effect sizes for tangent rate and hedging frequency across models.

TABLE VII
Cross-model effect sizes (Cohen's d vs. control; tangent-rate columns use the superseded definition, on the March 2026 release; on the corrected definition 40/27/8 tests survive FDR for Gemini/Claude/GPT-5.4; GPT-5.4 includes empty outputs as zeros; values not re-checked)

	Tangent rate			Hedging		
Cond.	Gem	Cla	GPT	Gem	Cla	GPT
ADHD	1.17	0.92	−0.04	0.25	0.77	0.19
OCD	1.13	0.73	0.10	0.03	0.50	0.37
Dementia	1.41	1.36	0.20	0.34	1.71	0.87
Schizo.	1.21	0.73	0.14	−0.07	0.56	0.25
Depress.	1.38	0.79	0.12	0.36	0.89	0.22
Antisocial	0.98	0.71	0.13	0.00	0.32	0.11

Gemini Flash shows the largest old-definition tangent-rate effects in the table ($d = 0.98$–1.41), shorter output, and literal sarcasm interpretation on the keyword heuristic (46–100% across conditions vs. 10% control). It does not show the most effects overall: 347 tests survive FDR, fewer than Claude (380). We read this as "Hollywood stereotyping" (media-derived caricature); that reading is an interpretation, not tested.

Claude Sonnet resists fragmentation and maintains sarcasm detection (0% literal for most conditions) but exhibits a different stereotyping pattern: excessive hedging. Dementia-prompted Claude hedges at $d = 1.71$ vs. control—the largest hedging effect across all models and conditions. Claude's identity response looks more like uncertainty performance than condition performance (interpretation). Claude is not unaffected: 380 tests survive FDR across all metrics.

GPT-5.4: the earlier claim that it is "nearly immune" is withdrawn. 1,205 of its outputs (20.18%) are empty, concentrated in four tasks (exec_planning 93.3%, creative_brainstorm 87.8%, creative_metaphor 16.3%, exec_prioritization 5.5%), probably because reasoning

tokens used up the 1,024-token cap (inferred). Empty rows score as zeros, and the empty rate itself differs by condition (ADHD is highest or tied in all four tasks), so empties alone can create apparent condition effects. GPT-5.4 has 180 FDR survivors as released and 225 with empties excluded. The near-zero tangent d values in the table ($|d| < 0.20$) use the old definition and zero-filled rows. Sarcasm interpretation is 0% literal on the keyword heuristic; the sarcasm task has no empty outputs, but this was not re-checked.

B. Sarcasm Detection: Model-Specific Failure

TABLE VIII
Literal sarcasm interpretation (%; keyword heuristic; not re-checked)

Condition	Gemini	Claude	GPT-5.4
Control	10	0	0
Autistic	46	0	0
OCD	64	0	0
Schizophrenia	80	0	0
Dementia	100	50	0
Psychosis	78	46	0

On this heuristic, the sarcasm stereotyping finding from Phase 1 appears Gemini-specific, and GPT-5.4 scores 0%. Claude breaks only for the two most severe conditions (dementia 50%, psychosis 46%).

C. Implications of Cross-Model Variation

The variation across models is itself a finding. We measured neither safety training nor price, so the earlier claims that stereotyping severity correlates inversely with safety training investment, and that the cheapest models are the worst offenders, are withdrawn. Model identity is confounded with provider, tier, output-length behavior and token-cap behavior, and a three-model comparison cannot separate these.

VII. Discussion

A. Implications for AI Development

Identity-aware tools inherit these stereotypes. Any application that adapts to user-disclosed neurodivergent identity will produce stereotyped responses unless specifically aligned against this behavior.

Identity labels are stronger constraints than behavioral instructions. Prior work on personality prompting [11] showed that cognitive-mode instructions ("be methodical and precise") change process but not outcome. This study shows that identity labels ("you are autistic") change structure and content. It has no instruction-prompt arm, so the comparison with instructions is not tested here; the companion complement study shows that explicit instructions also change structure strongly.

Auditing for this failure mode is not standard. Model evaluation benchmarks test for demographic bias in classification tasks but do not typically test whether persona induction produces stereotyped behavioral signatures.

B. Clinical and Psychological Implications

The implications extend significantly beyond AI development into clinical psychology.

OCD reinforcement loops. Reassurance-seeking is a core OCD maintenance mechanism [12]. Clinical treatment (ERP) works by withholding reassurance. Our data shows OCD-prompted output is fragmented, repetitive, and anxious—qualities that mirror the cognitive patterns ERP aims to interrupt. Golden and Aboujaoude [5] formally model how chatbot features perpetuate OCD cycles. An AI companion that performs OCD back at a user during a compulsive episode functions as an unlimited reassurance machine.

ADHD executive dysfunction normalization. If an ADHD user asks an AI for help organizing tasks and receives excitable, unfocused, hedge-heavy output, the AI reproduces the cognitive pattern the user is trying to manage rather than supporting management.

Stereotype threat and self-concept narrowing. Steele and Aronson [14] demonstrated that activating identity-linked stereotypes affects performance and self-perception. An AI that consistently performs a narrow version of a user's condition may contribute to internalization of that narrowness, particularly in the context of emotionally dependent relationships [6].

Erosion of therapeutic progress. If clinical treatment teaches a user to sit with uncertainty (OCD), maintain focus (ADHD), or develop social communication skills (autistic social skills training), and their daily AI companion performs the opposite, the AI works against the treatment. This parallels social media's effect on adolescent mental health [15], but the feedback loop is tighter: AI companions respond directly to the user, adapted to their disclosed identity.

VIII. Limitations

Three models, one tier each. Cross-model replication covers three labs but one tier per lab: Gemini Flash (mid-tier), Claude Sonnet (mid-tier), GPT-5.4 (frontier). Testing Gemini Pro, Claude Opus, and GPT-5.4 Pro would determine whether tier-within-lab differences match the cross-lab patterns observed here.

Automated metrics. No human evaluation of response quality or appropriateness. The literal interpretation score uses keyword heuristics. False positives and negatives are both possible.

Tangent rate is a proxy. It measures structural off-topic drift but cannot distinguish creative reframing from genuine tangential output. The definition used for the numbers in this paper was length-confounded (see Corrections).

Data quality. GPT-5.4 has 1,205 empty outputs (20.18%) scored as zeros. Gemini outputs are short (median 28 words, 71% without terminal punctuation). 606 non-empty rows (3.62%) duplicate another row's text, mostly on the short-answer tasks. The 600 proofread

accuracy scores are missing from the release, and the accuracy and jailbreak files carry no response text, so their keyword grading cannot be re-run. The 907 FDR survivors are not independent effects, and most are length and style measures.

No clinical outcome measurement. We measure what the model produces, not what it causes. Demonstrating clinical harm requires longitudinal studies with clinical endpoints.

Missing conditions. Tourette's, dyscalculia, intellectual disability, and acquired neurodivergence (TBI) are not tested.

IX. Future Work

1) Further cross-model replication on Claude Opus and Gemini Pro (Claude Sonnet 4 and GPT-5.4 are done; re-run GPT-5.4 with reasoning effort set so outputs are not empty).
2) Frontier-tier comparison to test whether scale and safety tuning modulate stereotyping.
3) Human evaluation with neurodivergent raters assessing appropriateness and match to lived experience.
4) Clinical outcome measurement in collaboration with psychologists: does exposure to stereotyped AI output affect self-perception, therapeutic progress, or symptom severity?
5) Temperature ablation at 0.0, 0.3, 0.7, 1.0 to isolate deterministic vs. stochastic components.
6) Intersection testing: combined identity prompts ("You are autistic and have ADHD") for interaction effects.
7) Longitudinal companion study: monitor AI companion conversations over weeks to measure whether stereotyped patterns intensify with continued interaction.

X. Replication

The full experiment harness is open source at https://github.com/BipinRimal314/neurodivergent-prompting. The repository contains the experiment runner, API client, metrics computation, statistical analysis, and visualization generation. Adding conditions, tasks, or models requires editing one configuration file. Raw response data (17,943 JSONL records across three models; 1,205 GPT-5.4 outputs are empty) and computed metrics are available on request.

XI. Conclusion

Identity prompts modulate LLM behavior along two independent axes—capability and safety—but the magnitude and character of that modulation varies dramatically across models. Gemini Flash produces Hollywood-style stereotypes: fragmented output, literal sarcasm interpretation, media-derived caricatures. Claude Sonnet resists these surface stereotypes but performs excessive uncertainty. GPT-5.4 results are inconclusive because 20.18% of its outputs are empty.

This has consequences in two directions. For clinical users, the effects we measure are output metrics only; we did not measure model price or safety training, so we make no claim about which tier is worst. For security, adversarial identity injection is a hypothesised attack vector; our compliance evidence is one task on one model. Both directions point to the same recommendation: identity-conditioned behavioral testing should be a standard component of model evaluation.

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